

soldiers and technicians to explore weaponry, machinery, automobiles, and other military equipment. This enables them to learn at their pace. The faster learner gets to take the tests earlier! Strategies can also be formulated better using interactive virtual platforms rather than presentations or charts.

We see start-ups coming up with innovative solutions for the defence sector—and the Indian military forces are quick to recognise, collaborate, and utilise these.

One such homogenous solution is the see-through armour developed by AjnaLens. The AjnaESAS (Enhanced Situational Awareness System) helps overcome a key problem faced by the crew inside a tank. It is very difficult to navigate a tank using the small periscopic view. AjnaESAS uses an AR based head-mounted display and a 360-degree camera to offer the crew a 360-degree horizontal field of view. The camera system also has night vision and 4X zoom capabilities. AjnaESAS uses AI to help navigate the tanks through challenging terrains by eliminating blind spots and providing clear images even when the view is obscured by dust or smoke.

Another of their solutions, AjnaBolt, improves weapon systems with features like thermal sensors for night vision, friend or foe detection (popularly known as IFF—identification, friend or foe), light detection and ranging (lidar), blue force tracking (a GPS-enabled system that gives a real-time picture of the battlefield, which conventional maps cannot give), and so on. Fair-weather weapons can be effectively upgraded to all-weather weapons, providing visual assistance in locating, locking, and engaging targets. When

it is combined with a holographic sandboxing system, it helps render a detailed 3D holographic map for planning and preparation. According to the company, “Users can interact

and annotate the maps using gesture inputs, enabling them to view the terrain from multiple perspectives and locate tactical advantages in the field. Further, the sandbox facilitates

remote networking capabilities, enabling effective coordination between the base and on-field operators.”

Another interesting product is the HoloSuit, manufactured in India by Mysuru based Kaaya Virtualisation Tech. The motion capture suit is embedded with over 40 sensors to capture the wearer’s movement from head to toe. The wearer’s movements are replayed in the virtual world, and the wearer also feels everything that happens there. For example, the user can operate a machine in the virtual world. The user also gets haptic feedback from the suit. If someone touches, shoves, punches, or shoots the digital twin, the wearer can feel it. It is being deployed by the Indian Army to train soldiers. According to news reports, the company has exported it to the South Korean Army as well, and it is also being used by one of the electricity boards in India to train workers to avoid electrical hazards.

Sports and fitness: Run to win

The metaverse will make sports and fitness much more interesting. We are not talking about immersive views of sports events, or trading of sports NFTs, we are talking about serious training and workout. For those of you who find the gym environment rather dull and boring but are unable to take out enough time to go rough it out in beautiful nature trails, the metaverse could be the perfect escape. Clip on your VR headset and you could be running through your favourite land-

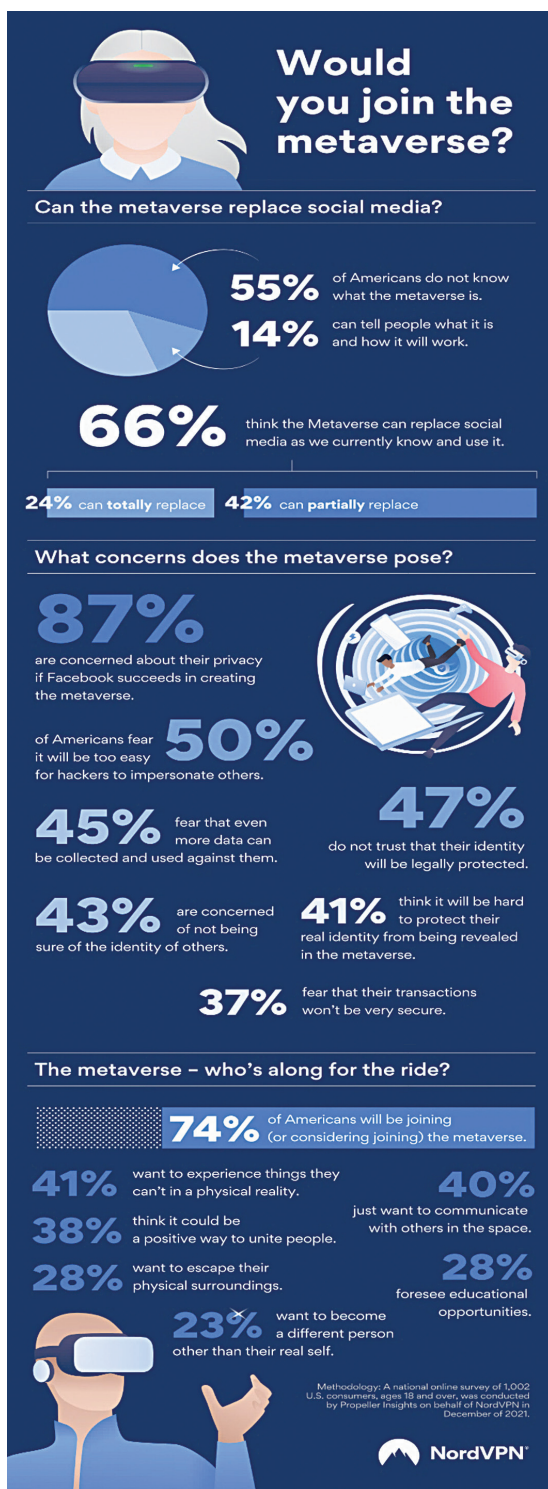


Fig. 4: A survey conducted by Propeller Insights for NordVPN in the US (Courtesy: NordVPN)